EXPERIMENT NO : 2

UPLOAD AND ANALYZE THE DATA SET GIVEN IN CSV FORMAT AND PERFORM DATA PREPROCESSING AND VISUALIZATION

AIM :

To upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

CODE :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv('sales_data.csv')
print(df.head())
print(df.isnull().sum())
df['Sales']=df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product','Quantity','Region'],inplace=True)
print(df.describe())
product_summary=df.groupby('Product')[['Sales','Quantity']].sum().reset_index()
print(product_summary)
plt.figure(figsize=(10,6))
plt.bar(product_summary['Product'],product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date']=pd.to_datetime(df['Date'])
sales_over_time=df.groupby('Date')['Sales'].sum().reset_index()
```

```
plt.figure(figsize=(10,6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()
pivot_table=df.pivot_table(values='Sales',index='Region',columns='Product',aggfunc='sum',fill_value=0)
print(pivot_table)
correlation_matrix=df.corr(numeric_only=True)
print(correlation_matrix)
plt.figure(figsize=(8,6))
sns.heatmap(correlation_matrix,annot=True,cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT :

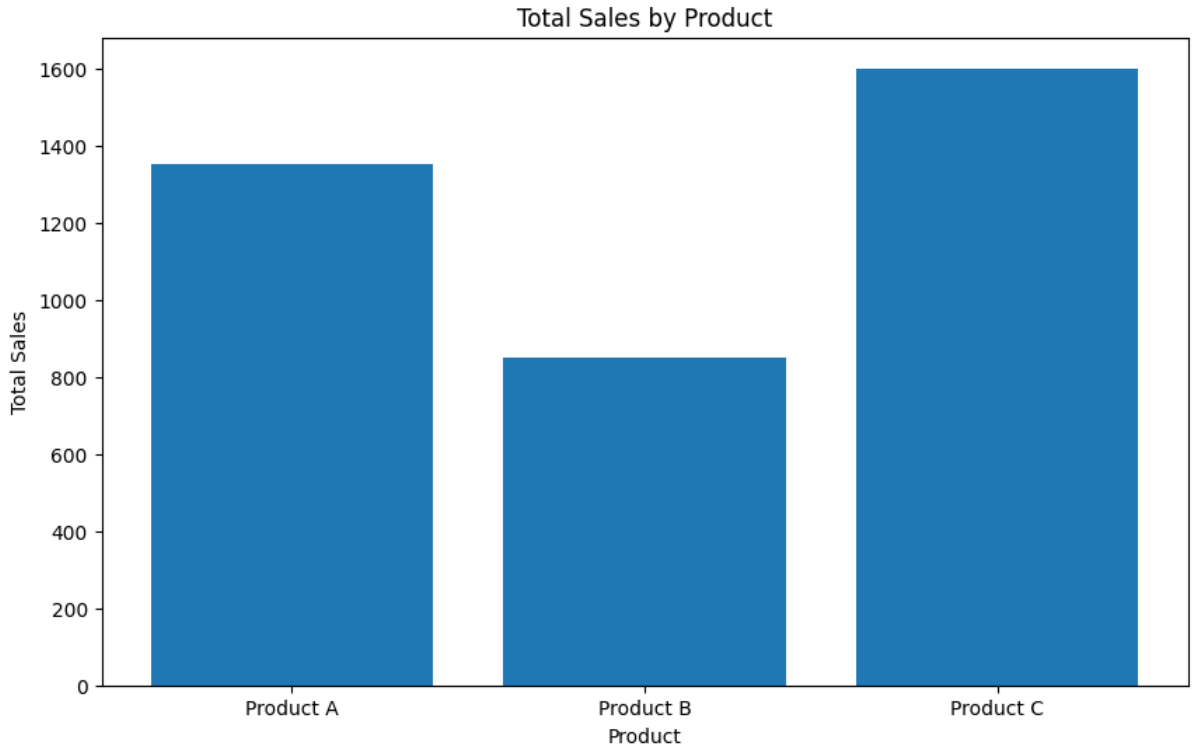
	Date	Product	Sales	Quantity	Region
0	2023-01-01	Product A	200	4	North
1	2023-01-02	Product B	150	3	South
2	2023-01-03	Product A	220	5	North
3	2023-01-04	Product C	300	6	East
4	2023-01-05	Product B	180	4	West

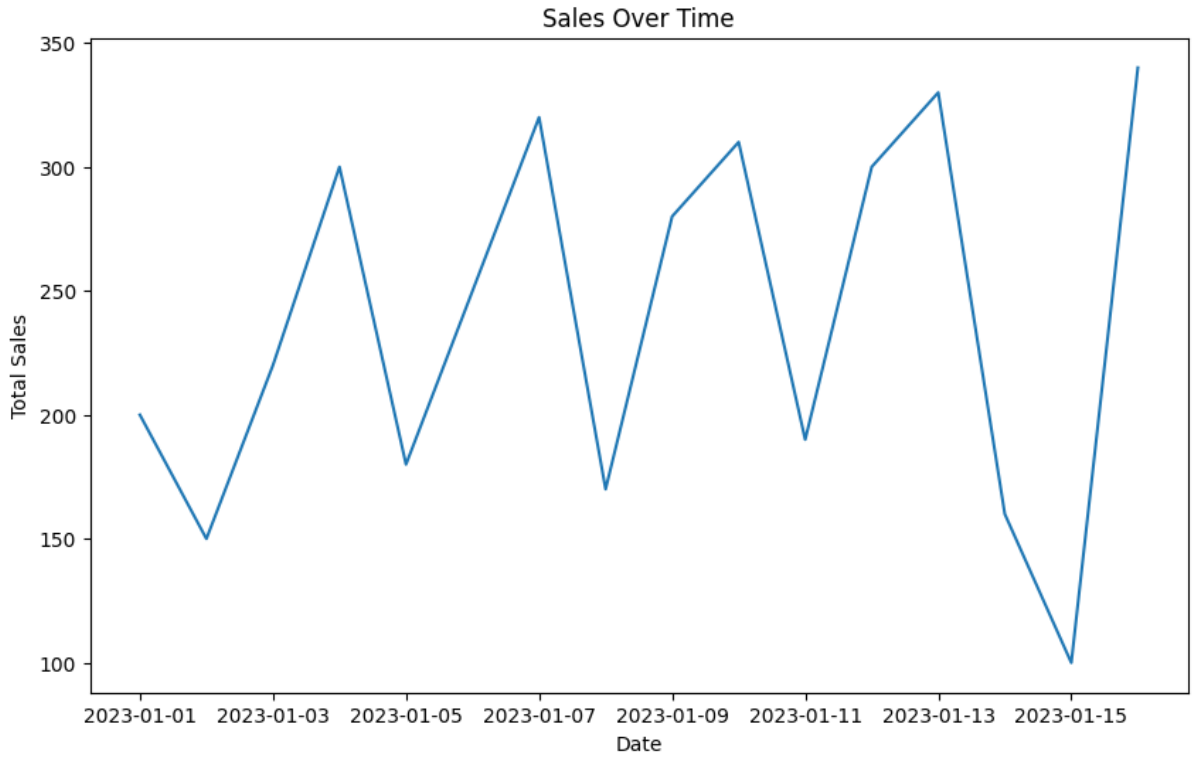
```
Date        0
Product     0
Sales       0
Quantity    0
Region      0
```

dtype: int64

	Sales	Quantity
count	16.00000	16.0000
mean	237.50000	5.4375
std	75.40999	1.7500
min	100.00000	3.0000
25%	177.50000	4.0000
50%	235.00000	5.5000
75%	302.50000	7.0000
max	340.00000	8.0000

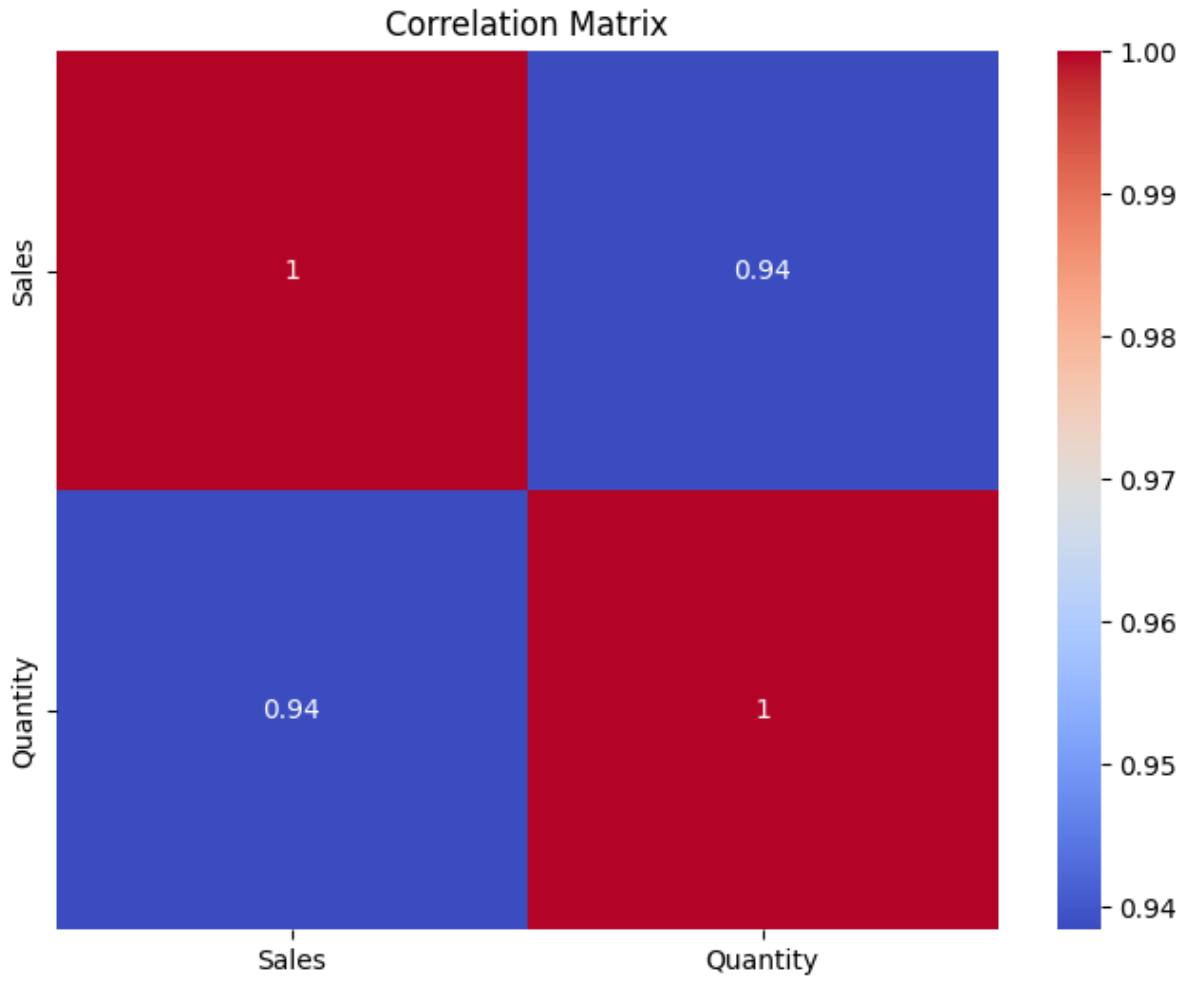
	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	18
2	Product C	1600	36





Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.938362
Quantity	0.938362	1.000000



RESULT :

Thus,the given data was analyzed successfully and required output was obtained.